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<nettime> CCTV/face recognition

Forced Entertainment - Tim on Wed, 19 Jun 2002 05:44:48 +0200 (CEST)

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<nettime> CCTV/face recognition

- *To*: <nettime-l@bbs.thing.net>
- *Subject*: <nettime> CCTV/face recognition
- *From*: "Forced Entertainment - Tim" <tim@forced.co.uk>
- *Date*: Tue, 18 Jun 2002 23:46:55 +0100
- *Reply-to*: "Forced Entertainment - Tim" <tim@forced.co.uk>

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a news story and a feature last week in The Guardian (UK) about face recognition software, CCTV surveillance and "exaggerated claims" by maufacturers Visionics/FaceIt. Esp beautiful explanation of the system's failure to spot a reporter whose face had been scanned for the system:

> "He passed our camera side on," said a spokesman for the borough, "so we were
> unable to detect him. This is obviously a weakness in the system."

tim

Towns secretly testing 'spy' software
James Meek, science correspondent
Guardian

Thursday June 13, 2002

A security system designed to automatically recognise people on the street and report their whereabouts to a central control room is being tested secretly in three places in Britain, according to the American manufacturer.

Three other areas - the London borough of Newham, Tameside in Greater Manchester, and Birmingham - are publicly using the system, known as FaceIt, made by Visionics.

A spokeswoman for the company, Frances Zelazny, said some councils had not

yet announced their use of the system because it was still "in the pilot phase".

The Home Office said it was up to local authorities whether they bought facial recognition software, which works as an extension of CCTV networks, and the government did not know which councils were using it.

The expansion of the FaceIt system in Britain comes amid growing doubts on both sides of the Atlantic as to whether the technology works. Civil liberties groups in the US claim Visionics has exaggerated the software's ability to pick out "known" faces in crowds.

In Newham, where the council claims non residential burglaries fell by 39% in the six months after FaceIt was installed in 1998, the local police officer responsible for providing photographs of ex-offenders to the council security service told the Guardian that FaceIt had never spotted a live target.

"There have never been more than 20 or 25 faces on the system," said Detective Inspector Ian Chiverton. "They've been weeded on a monthly basis. We've chosen what we call our nominal criminals, so they would be convicted burglars and rob bers, but only one facial shot was ever put on. And there's never been a recognition of that facial shot."

Visionics denied this. However, the council refused to provide details.

FaceIt failed to spot a Guardian reporter who voluntarily put his face into Newham's database and visited the areas covered by the cameras.

"He passed our camera side on," said a spokesman for the borough, "so we were unable to detect him. This is obviously a weakness in the system."

FEATURE:
http://www.guardian.co.uk/Archive/Article/0,4273,4432506,00.html

Robo cop

Some of Britain's 2.5 million CCTV cameras are being hooked up to a facial recognition system designed to identify known criminals. But does it work? James Meek has his mugshot taken, then hits the streets of east London - where they claim it reduces crime - to see if he gets spotted

James Meek
Guardian

Thursday June 13, 2002

I have a feeling that I am being watched by invisible eyes. Once this might have been interpreted as a symptom of mental illness. But just as those who jabber to themselves on the tube are now assumed to be talking on hands-free mobiles, my persecution complex of yesterday has become today's sane awareness of constant monitoring by CCTV cameras in public places. There is, however, a new stage to my mania. The They who are watching me, I believe, may not even be people. It may be a computer which knows what I look like, and when it sees me, it will recognise me, and sound the alarm.

This is not paranoia. Not entirely. It is London in 2002. Specifically, it is the east London borough of Newham, which has - with the backing of the Home Office - linked a computer to a small number of its hundreds of CCTV cameras and loaded the computer with a piece of software supposedly capable of recognising the faces of people on the street.

For three and a half years now, Newham council has been feeding the computer with photos of convicted local criminals at large: that is, people convicted of serious crimes who have either served time in prison and been released, or who were not jailed. Once programmed, so the theory goes, the computer becomes the perfect watchman. Dozens of CCTV cameras become its eyes. The eyes never blink. The computer never sleeps, nor does it get bored, nor does it miss the dodgy individual loitering near the ATM because it is ogling the girls at the bus stop. If the police are looking for someone, the camera will find them.

The police are not looking for me at the moment, but several weeks ago I voluntarily yielded a digital version of my face to a man from Newham council, who took it and fed it into the computer. I wanted to find out whether the borough's electronic watchers would be able to recognise someone and raise the alarm if they slipped into the detection zone unannounced. The answer spoke of a Blairite Britain less like George Orwell's 1984 than the world painted by Terry Gilliam's film Brazil, where secrecy, the commercialisation of public life and technologically advanced state surveillance are tempered by incompetence and machines that don't work.

Newham, a solidly Labour borough, is east East End, terrace after terrace of small brick houses interspersed with scruffy little shops, chain stores, pubs, preaching halls and mosques, a place whose inhabitants' collective memories reach back into every cranny of the history of the world's non-powerful people, in poor London, in Bangladesh, in Somalia, in Kosovo, in Afghanistan. Although it happens to contain London City airport within its boundaries, Newham didn't figure in the calculations of the world's powerful, until it got involved with Visionics. The US firm makes

the face recognition software, FaceIt, installed by Newham.

Visionics has trumpeted the alleged role of its software in reducing the borough's crime rate since 1998. After September 11, when Visionics' stock soared on Wall Street, the company made this little known London borough famous throughout America as the bold municipality which had dared to grasp the future of crimefighting technology, and won. Newham was Gotham City, Visionics was Batman, and though they never said it, the subtext was clear: let Osama Bin Laden dare to shop for yams in Stratford shopping mall, and - if MI5 had had the foresight to slip his image into Newham's database - he would be nabbed.

To get my picture taken I travelled to the council's Folkestone Road depot, where, behind multiple sets of security doors, shifts of operators spend their days and nights watching banks of screens showing what must be one of the least interesting works of reality TV ever devised, the unedited, real-time, street life of Newham. Despite the apparent dullness of their job, there is an air of bustle and commitment among the uniformed staff, much banter and tea, reinforcing the sense that, given a uniform, a computer and access to a kettle, there is nothing a British guard won't do.

Newham was a convert to the virtues of CCTV even before Visionics came along. When the two went into partnership Newham already had more than 140 street cameras and 11 mobile camera units. "I'm not that uneasy about surveying people on cameras because the world is full of cameras and public opinion in Newham says over 90% want more cameras, not less. This is east London and their attitude is, if you're doing nothing wrong, you've got nothing to worry about," said John Page, Newham's head of emergency services and community safety.

John Tisshaw, who carries the title of senior enforcement officer, said: "Since 1996 there's been a big call for more CCTV, not less. It reduces fear of crime, makes people's life seem very much better. As a result, everybody wants cameras, apart from those few people who want to cause criminal offences."

I wasn't the first to try to take the facial recognition challenge. John Cleese had been, wearing a dress. So had a reporter from the US network CBS. "We got him every time except once where he was wearing a false beard and glasses," said Tisshaw. "But he said that everybody was looking at him at that point."

Already at Folkestone Road there were signs that FaceIt was not all it was cracked up to be. Page and Tisshaw were strangely reluctant to say how many times the police had reacted to a Visionics alarm. Indeed, Tisshaw hinted that the true value of FaceIt was in deterrence, regardless of

whether it worked or not, like a fake burglar alarm on the outside of a house. "The reason we don't publish these figures is because it's just contrary to the way we run this scheme. What we're saying to criminals is: this is the system we've got, and we don't want you doing your wrongdoing, because we know you're around."

After Newham put my picture on their villains' database - a single face-on shot taken with a regular digital camera - we agreed that I would test the system by visiting one or more of the zones covered by FaceIt. I wouldn't tell them when I was coming, but it would be within an agreed seven-day period. That period took a long time to arrange. So long that, while I was waiting, I happened to visit the borough for an unrelated reason. I was wearing an off-white linen suit. I think I was the only one in Newham wearing an off-white linen suit that day. From the point of view of the security cameras, I might as well have been wearing a sandwich board with the words "Villain - sample only" on it. But the software was being tinkered with that day, so the computer had no chance.

In early May, I got the call from Newham: they were ready. I picked Friday afternoon for my raid. I thought about false facial hair. I thought about cotton wool in my cheeks and dark glasses, and wigs. In the end I didn't try to disguise myself but I did wear a baseball cap, which I hadn't had on when the picture was taken. At first acquaintance it might seem that Visionics has solved a problem common to all camera-based surveillance systems, the Big Brother problem: if Big Brother is watching you, who is watching everybody else?

Suppose a security organisation wants to track, in real time, the movements of 100 people as they pass through a city. Suppose it has camera coverage of the entire area. To be certain of keeping its human targets in view at all times, it needs one watcher per camera. The watchers need screen breaks. There needs to be at least two shifts. That means several times more watchers than targets, hundreds of watchers, even without anyone to manage the system, let alone go out and pick up suspects.

Visionics claims its system can get round this by automatically recognising people's faces, distinguishing suspects under surveillance from other people, even on a crowded street. Here is how it is supposed to work: when the computer receives the two-dimensional camera image, it sorts the faces from the rest of the background clutter and, for each face, picks out a series of "nodal points" and measures the distance between them. Visionics has worked out 80 nodal points - "the peaks and valleys of the bone structure of the face", as a company spokesman put it. They lie along the edge of the eye socket, the jawline, the nose, anywhere that marks a clear boundary between light and shade. The distances between points are converted to a string of numbers which are then compared against the set of suspects' face numbers in the database. If there is an

80% chance of the match being right, the computer alerts an operator in the CCTV control room and displays the two facial images on screen for a human check. If the operator confirms that a suspect has been spotted, he contacts the police, who decide whether or not to take action. As few as 14 nodal points are enough for a good match.

If you are one of those responsible for catching criminals, or indeed monitoring political dissenters, this might sound good. But there are problems. What if targets disguise themselves more expertly than John Cleese or the CBS guy? Where do you get the pictures on your criminal database from in the first place? Convicted criminals? But if they are convicted, why are they walking the streets? If they are harmless enough to be bailed, or have served their time, why are computers spying on them? And what are the police supposed to do when they get the call that there is a known ex-con walking through the shopping centre? Send a response team? Walking through a shopping centre is not a crime, even for former criminals.

It turns out that there is an even deeper problem with the Visionics software in what the company calls "ambient surveillance" situations. It doesn't work. There is certainly no proof that it ever has in Britain. I learned that from someone who should know, Detective Inspector Ian Chiverton, the police officer responsible for liaising with Newham council on the Visionics package. I called on him before I set off for the FaceIt hot zones.

'There have never been more than 20 or 25 faces on the system," he said. "They've been weeded on a monthly basis. We've chosen what we call our nominal criminals, so they would be convicted burglars and robbers, but only one facial shot [of each] was ever put on. And there has never been a recognition of that facial shot."

That is right, never. Not once, as far as the police know, has Newham's automatic facial recognition system spotted a live target. It is not surprising. DI Chiverton explained that for the system to work properly, it needs pictures taken of suspects from at least five different angles and the Metropolitan Police doesn't take that many. "To be recognised you would have to walk around Newham like this, wouldn't you," he said. He got up from his desk and stomped around the room, holding his face in a grinning rictus at a 45 degree angle to the ground. "It's not recognising your face - it's a computer, isn't it? It recognises digital imagery - how many times are you going to walk around Newham like that?"

Chiverton wasn't saying the system might not work one day, just that neither it, nor the police resources to make use of it, were ready. He could see it being useful for tracking people on the sex offenders register, or to hunt for missing persons. "If it catches one murder

suspect that would be worth it," he said. But with Newham police 80 to 90 officers down on full strength, and anxious not to become embroiled in new human rights or civil liberties battles, there was little they could do with the system as it stood, even if it worked.

"Say you're Bill the burglar and I was just to get your face on the system and a little bell rings at Folkestone Road - what would I do with it? I haven't got the resources to follow you around Newham. So what is the purpose of putting 300 Bill Burglars on there? It's different if you're wanted for murder or you're a paedophile. But then you have to make sure you have the management systems in place."

I left DI Chiverton and set off into deep Newham to test the cameras. There are few zones in the borough where the cameras are linked to FaceIt, and I promised council officials that I wouldn't reveal where they were, but it is not giving much away to say that they are places where there is a great deal of coming and going. I visited two of them. I hadn't intended to make it easy for the system. I tried to think like a man who does not want to be seen, who wants to blend in with the crowd; swift, anonymous, the assassin, the bomber, the pickpocket. I blew it. As I approached the hot zones, my heart began to pound and I began to sweat. I had a sense that I was being tracked. I also had a genuine curiosity to see the cameras, so I began looking up, like a tourist in Manhattan, then remembered that if I could see the cameras they would see me, so I looked down at the ground. At this point I was almost run down by a family in a mini-van. I was about to leave the area, then remembered I had meant to get a timed and dated receipt to prove I had been there, so doubled back, then back again; then I couldn't find my travel ticket and spent five minutes going through 10 pockets with my face to a wall. I was relieved that no human passer-by, let alone a computer, had called the police because of this dodgy behaviour.

I needn't have worried. FaceIt didn't spot me, in that zone or in the second one, where there had been just as much palaver, and I took off my hat. The computer never knew I had been there. Sandra, the always cheerful Newham press officer, didn't sound surprised when she reported the failure of the Visionics system, and by this time, neither was I.

The vaunted clear-cut success of FaceIt in reducing crime in Newham more than in neighbouring boroughs doesn't stand up to close analysis. True, between 1999 and 2002, crimes of violence in Newham went up by 4%, compared to 19% in the neighbouring borough of Tower Hamlets, and the fall in criminal damage in Newham - 8% - was greater than the 1% fall in Tower Hamlets. But theft increased by about the same - 11%in the latter, against 10% in Newham. Most remarkably, whereas robberies in Tower Hamlets went up by 33%, in Newham they increased by a still more alarming 38%; and while in Tower Hamlets burglaries fell over the period by 12%, in Newham, they

went up slightly.

Another blow to Visionics' credibility in street surveillance came from the US, where, using open government legislation unavailable in the UK, the American Civil Liberties Union forced police in Tampa, Florida, to make available daily logs of the FaceIt system they installed last year. The logs showed that on four days in July, the system had sounded the alarm 14 times. This sounds better than Newham. It wasn't. Every one of those 14 turned out to be a false match - an innocent member of the public, unwittingly thrown up for scrutiny by a security service.

The man who knows most about Newham's love affair with facial recognition technology is Bob Lack, Tisshaw's predecessor, who oversaw the system's introduction and was present when Home Office officials carried out an assessment. Unfortunately, several months ago, something happened to Bob Lack which means that it would be easier to talk to him if he was living in a shed on the outskirts of Ulan Bator. He was given a job at the Home Office. Lack told me that he couldn't speak to me without permission from the government press office, and the government press office told me: "We don't facilitate interviews with officials."

Clive Reedman of the Police Information Technology Organisation seemed more helpful. This was an illusion. "I can't give you figures on the numbers of times the alarm sounds [in Newham]," he said. "Even if the alarm does sound it doesn't necessarily mean the police are going to take any action because it's an operational decision. It is an extremely difficult thing to evaluate from the technical point of view. The best you can say is that it's a successful system, the main reason probably being deterrence."

How did he know it was successful? "The system has been evaluated. What I can't give you is any results, because that is between us and the Home Office. What I can say is, you have to remember a face-in-the-crowd system is extremely difficult to evaluate anyway.'

But surely the public had a right to know whether the system worked or not? No, said Reedman. "The figures, if released, would never be viewed in the right way. It is very, very difficult to define a standard trial for a face to face system. So there is an awful lot of subjectivity, open to all kinds of interpretation. All we can say is, the Home Office is happy in the investment in the [Newham] CCTV system because it has had the purpose of reducing crime and removing the fear of crime.'

Certainly the Home Office does seem to be happy. Extraordinarily happy, as it turned out when I finally called Visionics. So far, only two councils in the UK have gone public with their use of FaceIt - Newham and Birmingham. According to a Visionics spokeswoman, Frances Zelazny,

however, four other local authorities in Britain are running trials of the system. She just couldn't say which ones because they weren't ready to go public.

I asked about a press release Visionics put out on January 10, saying that "the FaceIt surveillance system, running in a UK town centre, successfully identified a subject wanted by law enforcement authorities". Could she say where this happened and what kind of suspect was involved? She couldn't. I asked about DI Chiverton's denial that the system had ever spotted a suspect. Zelzany rejected this. 'There was a Home Office official in the Newham control room during August 2001, to evaluate the system, and they found that of about half a million faces, 93 were referred to the police as matches. It wasn't up to Bob Lack whether the police took it seriously."

But if the Home Office wouldn't release the results of the evaluation, and Lack wasn't allowed to talk, and the police said there had never been a good match, didn't that suggest the system didn't work for street surveillance?

"People who use technology in these areas understand that it's a tool that helps them, not the be all and end all," said Zelzany. "[FaceIt] is doing something it's virtually impossible for humans to do. There's no way a human can sit in front of 500 screens and remember hundreds of faces and match them all."

There are about 2.5 million CCTV cameras in Britain, 10% of the worldwide total. An unknown number of them, in unknown places, are being hooked up to a computerised facial recognition system which, on the available evidence, doesn't work. This might seem absurd, rather than sinister, where it not for the fact that the process is being carried on either in secrecy or, as in Newham, behind a blare of publicity which conceals the lack of real information about how well the system works. You have been warned. But you haven't been told. Smile!

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